

HENIL PATEL

ELECTRONICS & COMMUNICATION ENGINEERING STUDENT

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SUMMARY

Electronics and Communication Engineering student with hands-on experience in FPGA-based digital system design using Verilog HDL. Developed hardware projects including a 16-bit ALU with status flags, real-time timer systems, and FPGA-based interactive applications. Strong foundation in digital logic, RTL design, and hardware interfacing. Interested in advancing in VLSI/FPGA domain through research-oriented work and practical hardware implementation

TECHNICAL SKILLS

- Hardware & Design: Verilog HDL, Digital Logic Design, FPGA (Quartus, ModelSim)
- Embedded Systems: Arduino, ESP8266, Sensor Interfacing
- Programming: C, C++, Python
- Tools: VS Code, Arduino IDE

PROJECTS

16-bit ALU with Status Flags using Verilog (FPGA Implementation)

- Designed ALU supporting arithmetic and logical operations with status flags (CF, ZF, OF, SF)
- Verified functionality using RTL simulation and implemented on FPGA hardware
- Used control signals for operation selection and real-time output display

FPGA-Based Whack-a-Mole Game

- Developed real-time interactive system using Verilog with timers, counters, and LFSR
- Implemented modular architecture (timer, random LED generator, scoring system)
- Integrated LEDs, switches, 7-segment display, and LCD interface

FPGA Car Lap Timer System

- Designed multi-level timing system (milliseconds, seconds, minutes)
- Implemented clock division and lap-time storage using registers
- Displayed real-time and stored outputs using 7-segment display and LCD

EDUCATION

B.Tech in Electronics & Communication Engineering

Institute of Technology, Nirma University | 2025 – Present

Diploma in Computer Engineering

Government Polytechnic, Gandhinagar (GTU) | 2022 – 2025

CGPA: 8.99

RELEVANT EXPERIENCE

Embedded Systems Intern – AICTE IDEA Lab (GTU)

- Aug 2023

Python & Django Intern – STYPIX

- Jun 2024 – Aug 2024

Artificial Intelligence Intern – Sahana Systems Ltd.

- Jun 2024 – Aug 2024

RELEVANT COURSEWORK

Digital Logic Design, Electronics Devices, Signals & Systems, Communication Systems, FPGA-Based System Design